Software Requirements Specification for Traffic Violation Reporting System

Version 1.0 approved

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Table of Contents

Revision History
1. Introduction
1.1 Purpose
1.2 Document Conventions
1.3 Intended Audience and Reading Suggestions
1.4 Project Scope
1.5 References
2. Overall Description
2.1 Product Perspective
2.2 Product Features
2.3 User Classes and Characteristics
2.4 Operating Environment
2.5 Design and Implementation Constraints
2.6 User Documentation
2.7 Assumptions and Dependencies
3. System Features
3.1 System Feature 1
3.2 System Feature 2 (and so on)
4. External Interface Requirements
4.1 User Interfaces
4.2 Hardware Interfaces
4.3 Software Interfaces
4.4 Communications Interfaces
5. Other Nonfunctional Requirements
5.1 Performance Requirements
5.2 Safety Requirements
5.3 Security Requirements
5.4 Software Quality Attributes
6. Other Requirements
Appendix A: Glossary
Appendix B: Analysis Models
Appendix C: Issues List

Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The Traffic Violation Reporting System is a software application exclusively dedicated to reporting traffic violations. It collects information about the traffic violators along with the required documentation. This formally documents the agreement between the project team and stakeholders regarding what the system will do, how it will behave, and under what constraints it will operate.

This document is planned to give a complete and clear understanding of the system to developers, instructors, evaluators, and future maintainers. It broadly specifies the scope, features, functionality of the user, data usage, and the attribute quality of the system. The document will serve as the main reference in the design, development, testing and evaluation stages of this project.

The Traffic Violation Reporting System helps to allow the citizens to report traffic violations using evidence such as photos or videos, gives the power to traffic authorities to verify those reports and helps in the generation and management of challans. Apart from that, features like getting vehicle details through number plates, getting details of challans, tracking the validity of vehicle documents (insurance, pollution certificate etc.), reward-based incentive mechanism for valid reporting will also be available.

This SRS covers the complete system including:

- User-side functionalities (citizen reporting, challan viewing, rewards, vehicle details)
- Authority-side functionalities (verification, challan issuance, report management)
- Administrative controls (user management, system monitoring)
- System-level services (security, validation, notifications, and data storage)

1.2 Document Conventions

This SRS document conforms to software engineering documentation standards to ensure readability, consistency, and clarity.
All requirements are clear, concise and easily understandable. All of the functional requirements must be written using “The system shall…” for verifiability and testability.
The following conventions are used:

- Section headings and subheadings are numbered for easy navigation.
- Functional requirements are uniquely identified using IDs such as REQ-1, REQ-2, etc.
- Priority levels are expressed as:
- High – Mandatory for basic system operation
- Medium – Important but not critical
- Low – Optional or future enhancement

- Bullet points are used for listing features and behaviors.
- Technical terms are defined in the Glossary (Appendix A).
- Abbreviations such as SRS, GPS, RC, UI, and HTTPS are used consistently.
- "TBD" (To Be Determined) may be used where exact values or decisions are pending.

1.3 Intended Audience and Reading Suggestions

The purpose of this document is for the project instructor, evaluators, developers, testers and project team. It offers a detailed description of the Traffic Violation Reporting System so that stakeholders have a common understanding of the system objectives and functions. This document will serve as a reference for both developers and testers. During design and implementation the developers will refer this document. The testers will use this document to check whether the working of the system is in accordance with the requirements. Highly Approved by Evaluators and Instructors to Evaluate Correctness, Completeness, and Feasibility of the Proposed System.

New readers of the project should read first the Introduction and Overall Description. This will give them an overall idea of what the system is. You will read Section 3 (System Features) after that to understand its detailed functionality. If you're interested in the technical interfaces and quality requirements, sections 4 and 5 are the most relevant. The appendices provide supporting information, including definitions, models and outstanding issues.

1.4 Project Scope

The Traffic Violation Reporting System offers a web-based platform for reporting, validating, and managing traffic offenses in an easy and transparent manner. Citizens can use this system to send evidence of traffic violation in the form of photos and videos, obtain vehicle information using number plate details, track the status of their complaint, view challans and check the validity of vehicle like insurance, pollution certificate etc. It also offers citizens a rewards mechanism for responsible reporting.

Through this system, the traffic police authorities can help to verify the reports submitted, approve or reject evidence, and so on. Also, they can raise a challan for any violation that is valid. The administrative module serves to manage users, regulate system operations and monitor performance. The aim of the platform is to utilize technology to reduce manual paperwork, improve accountability, and increase public participation.

This project is aimed at providing only digital support for reporting and management. System excludes any legal enforcement actions, physical traffic control measures and final decisions of the judiciary. In the future, users may use their earned cashbacks to pay challans and get even more connected with government databases.

1.5 References

This Software Requirements Specification has been prepared by following standard software engineering documentation practices and guidelines. The structure and organization of this document are inspired by the IEEE Recommended Practice for Software Requirements Specifications (IEEE 830 / ISO/IEC/IEEE 29148), which provides a standard format for defining system requirements clearly and systematically.

Traffic regulation context and terminology referenced in this document are based on general guidelines provided by the Ministry of Road Transport and Highways (MoRTH), Government of India, and commonly followed traffic enforcement practices.

2. Overall Description

2.1 Product Perspective

The Traffic Violation Reporting System is a new stand-alone software product that supports and improves the existing traffic law enforcement process digitally. While it does not replace current systems of legal enforcement and physical enforcement, it will support them by improving reporting, verification record management and change management.

The platform connects citizens with traffic authority so that citizens can report any traffic violations by an easy reporting structured system where officials can view and act accordingly. For a long time now, traffic violators are identified through the manual observation of traffic police, physical challans on the vehicles, and offline record keeping. The digitalization of system (which means its transformation into a computer-readable form) reduces the paper work. Also it makes the system more transparent,efficient and accountable.

- A user interface layer for citizens, traffic police, and administrators.
- An application layer that handles business logic such as report validation, challan generation, and reward calculation.
- A data layer that stores user information, violation records, evidence files, and transaction logs.

The Traffic Violation Reporting System is built for scalability, allowing for future enhancements such as online challan payment, artificial intelligence-based violation detection, and more integration with government systems, all of which can be added without altering the main structure of the system.

2.2 Product Features

The Traffic Violation Reporting System comprises of a suite of integrated features enabling citizen and traffic authorities interaction through a single digital platform. The system allows citizens to report traffic violations by submitting a photo or video as well as essential details such as location and timing of the incident. The system can get and display information related to the vehicle using the vehicle number plate for accurate identification.

The users can view the challan issued to them, check the status of the lodged violations, and check the validity of vehicle documents including insurances, pollution, and registration. A reward mechanism has been created to motivate people to report violations responsibly, giving the first person who properly reports a cashback or points. In future updates, this cashback you earned will be applicable for paying challans through the system.

The system provides traffic police with tools to review the submitted report, authenticate evidence, approve or reject the violations and issue the challan in confirmed cases. It has a digitally accurate record of everything done which is open to scrutiny.

The admin module allows system administrators to create users, control access permissions, view system performance and send analytical reports. It also maintains freshness and stability of the system by handling misuse and suspicious activities.

The system is also equipped with core support features including user identity security, automatic timestamp and location, duplicate report detection, evidence format verification, and notification services. Together these features make sure the platform is reliable, secure and practical to be used in the real world.

2.3 User Classes and Characteristics

Citizen/User is the First User Class This refers to any registered person using the system to report traffic violations or check information related to his vehicle and those challans. Citizens are expected to know the basics of a smartphone or web application. Users will have to do the basic registration and signing in to the system, upload any photo or video of the traffic violation, view the status of the report uploaded, check on the issued challans, and also check on the validity of the documents of the vehicle along with earning reward points for every valid report. It is not in the purview of citizens to verify violations or issue challans.

Traffic police officer is the second user class. The authorized personnel of the traffic department verifies the reported violations in this user class and acts when needed. Traffic police will be expected to have Average technical knowledge and familiarity with official traffic rules and

procedures. Responsibilities include scrutinizing received reports, validating existing evidence, accepting/dismissing violations, issuing challans, and accurately documenting them digitally. The can't edit system settings and manage users beyond what is assigned to them.

The Administrator is the third user class. System administrators' access is the highest, and they manage the overall system. Technical knowledge and strong understanding of system operations is expected of them. The responsibilities include creating traffic police user accounts, monitoring activity in the system, handling user-related complaints/issues in the system, managing security control, report generation, and ensuring system stability. Admin is not involved in violation verification and challan making, but they ensure the platform is running smoothly and securely.

2.4 Operating Environment

The Traffic Violation Reporting System will work in a web-based and/or mobile-based environment. The system offers access to users through web browsers and mobile applications equipped in smart devices like mobile phones, tablets, or computers. The system is designed to be platform independent so that it can function on different operating systems without any limitation.

The system's server side will run under a centralized server environment that will manage database, host applications, and store data securely. A stable internet connection will be required for real-time communication between users and the system. All the information related to a user, violation, evidence, challan, reward, and system logs will be stored in the database.

The software will be easily installable on the windows, Linux, Android and other compatible systems for development and deployment. The client side should compatible with modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, mobile platform Android, iOS, etc.

To improve accuracy, the system may use external services such as GPS for location tracking, camera access for taking evidence, and number plate information services for vehicle identification. This system should not interfere with the normal functioning of any of these services.

The environment in general is designed to be flexible and scalable and access to the system so that it can be used effectively by citizens, traffic police and administrators in real conditions.

2.5 Design and Implementation Constraints

The Traffic Violation Reporting System should be designed with common and trustworthy technologies to ensure ease of development, maintenance, and enhancement. It should be

compatible with common web and mobile platforms without relying on specialized or proprietary software. A stable internet connection is necessary for reporting, verification, and data retrieval.

The uploaded images and videos should have size and format constraints to ensure storage space management and system performance. Differences in device capabilities, such as camera resolution and GPS precision, can impact evidence quality and should be taken into consideration during development.

Integration with external services such as vehicle registration and document verification systems is dependent on the availability of official APIs. In an academic or testing setup, sample or simulated data sources can be employed instead of actual government databases.

Security is a hard constraint. The system should provide secure authentication, role-based access control, and encrypted data transfer to safeguard personal and vehicle details.

Because of time and resource constraints in an academic setting, only basic functionality will be included in the first release. More advanced features like online challan payment, AI-assisted violation detection, and automated fraud prevention will be considered later.

2.6 User Documentation

The Traffic Violation Reporting System will be accompanied by basic user documentation to facilitate different users in comprehending and utilizing the system effectively. This will include a simple user manual that will guide users on how to register, log in, report traffic violations, upload evidence, view challans, check vehicle information, and monitor the status of their reports. Short user manuals will also be provided for traffic police officers on how to authenticate reports, accept or reject traffic violations, and generate challans. An administrator manual will outline system management operations such as user management and monitoring of activities.

Additionally, the system will incorporate on-screen help, tooltips, and error messages to assist users in the course of its normal usage. Given that the system is user-friendly, the documentation will not be required to focus on the technical aspects of the system. The documentation will be in digital form and can be modified as new functionalities are incorporated into the system.

2.7 Assumptions and Dependencies

It is assumed that the users have access to a smartphone or computer with a stable internet connection. The citizens are expected to provide genuine and clear evidence while reporting any violation. The traffic police officers are assumed to check the reports responsibly as per legal procedures.

The system relies on the availability of basic services such as GPS, camera connectivity, and proper data storage. The connectivity with the vehicle number plate databases and document verification systems rely on the availability of official APIs. If these services are not available, sample or mock data can be used for academic purposes.
The system relies on proper server setup, database stability, and security features. Any change in legal regulations, API availability, or resources may require modifications in the system requirements.

3. System Features

3.1.1 Description and Priority

This functionality enables users to register and log in to the system securely. It also ensures that only authorized users are able to use the system's functionality.
Priority: High

3.1.2 Stimulus/Response Sequence

- User registers by opening the registration page and submitting information → system registers the user.
- User attempts to log in by submitting login information → system authenticates and logs the user in.
- User provides incorrect login information → system prompts an error message.

3.1.3 Functional Requirements

REQ-1: The system must enable users to register by providing name, phone/email, and password.
REQ-2: The system must check the required fields before registering.
REQ-3: The system must enable registered users to log in securely.
REQ-4: The system must prompt an error message for incorrect login information.
REQ-5: The system must retain the user session after a successful login.
REQ-6: The system must enable users to log out securely.

3.2 Traffic Violation Reporting

3.2.1 Description and Priority

This service allows citizens to report traffic violations by providing evidence and details of the incident.
Priority: High

3.2.2 Stimulus/Response Sequences

User uploads photo/video and enters vehicle number → system records report.
Missing required fields → system asks user to fill in the details.

Report submitted successfully → system provides report ID and status.

3.2.3 Functional Requirements

REQ-7: The system shall provide functionality for users to upload an image of the violation.
REQ-8: The system shall provide functionality for users to enter vehicle number details.
REQ-9: The system shall automatically record the date and time of submission.
REQ-10: The system shall automatically generate a unique report ID.
REQ-11: The system shall automatically assign the report status as "Pending" after submission.
REQ-12: The system shall prevent users from uploading files with unsupported formats.

3.3 Violation Verification and Challan Issuance

3.3.1 Description and Priority

This functionality will enable the traffic police to check the violation reports and generate challans for the valid ones.
Priority: High

3.3.2 Stimulus/Response Sequences

Officer checks report → approves or rejects it.
If approved → challan is generated.
If rejected → report status is updated.

3.3.3 Functional Requirements

REQ-13: The system shall enable authorized officers to check the submitted reports.
REQ-14: The system shall enable officers to approve or reject reports.
REQ-15: The system shall enable officers to input challan amount.
REQ-16: The system shall update report status after verification.
REQ-17: The system shall record all verification activities in the form of records.
REQ-18: The system shall show the updated status to the user who submitted the report.

4. External Interface Requirements

4.1 User Interfaces

The system provides a simple, intuitive, and user-friendly interface designed for ease of navigation and accessibility. The interface ensures that users with basic technical knowledge can operate the system efficiently.
The citizen interface includes registration and login pages, a dashboard for reporting violations, evidence upload options, and sections to view challan status, reward points, and vehicle information. Traffic police users are provided with a verification dashboard where submitted reports can be

reviewed, approved, or rejected. The administrator interface includes controls for managing users, monitoring reports, and maintaining system records.
The interface will include clear input forms, status indicators, and error messages to guide users during interaction. The design emphasizes readability, clarity, and minimal complexity to ensure smooth usability.

User interface features include:

- Registration and secure login pages
- Dashboard for reporting and tracking violations
- Evidence upload interface with preview option
- Verification panel for traffic authorities
- Administrative control panel
- Clear notifications and error messages

4.2 Hardware Interfaces

The system interacts with standard user devices but does not require specialized hardware. Citizens may use smartphone or computer cameras to capture images of traffic violations before uploading them to the system.

The application relies on standard device capabilities such as camera access, storage access, and internet connectivity. Since the system operates through web browsers, no direct hardware control is required.

Hardware interaction includes:

- Smartphone or device camera for capturing violation evidence
- Device storage for selecting and uploading media files
- Standard computing devices (mobile phones, tablets, laptops, desktops)
- Internet connectivity hardware for data transmission

4.3 Software Interfaces

The Traffic Violation Reporting System interacts with backend software components and database systems to store and retrieve user data, violation reports, challan records, and reward details.

For the academic version of the project, vehicle information and document validity details will be retrieved from a local database. Future versions may integrate external services such as vehicle registration systems or insurance verification APIs.

The system is designed to be compatible with modern web browsers and standard operating systems, ensuring cross-platform accessibility.

Software interactions include:

- Database management system for storing records
- Server-side application logic for processing reports
- Web browser compatibility for user access
- Future integration capability with external APIs

4.4 Communications Interfaces

The system uses internet-based communication protocols to enable data exchange between users and the server. Secure communication ensures that personal information and violation data remain protected during transmission.

All user actions such as report submission, verification updates, and status retrieval depend on network connectivity. The system is designed to function efficiently under typical internet conditions.

Communication features include:

- HTTP/HTTPS protocols for secure data transfer
- Internet connectivity for real-time access
- Secure transmission of user and report data
- Server-client communication for status updates

5. Other Nonfunctional Requirements

Nonfunctional requirements define the quality attributes and operational standards that the Traffic Violation Reporting System must satisfy. These requirements ensure that the system performs reliably, remains secure, and delivers a consistent user experience. While functional requirements

describe what the system does, nonfunctional requirements specify how well the system performs under various conditions.

The system is designed to provide dependable performance, data protection, usability, and maintainability. These attributes ensure that the platform remains efficient and trustworthy for citizens, traffic authorities, and administrators.

5.1 Performance Requirements

The system should provide fast and responsive performance during normal operation. Users should be able to submit violation reports, upload evidence, and view status updates without noticeable delay. The system must be capable of handling multiple users simultaneously without performance degradation.

Efficient database operations and optimized media handling will ensure smooth processing of reports and retrieval of records. Performance should remain stable under moderate usage conditions typical of an academic demonstration system.

Performance requirements include:

- Quick response time for report submission and status retrieval
- Ability to support multiple users at the same time
- Efficient storage and retrieval of records
- Optimized handling of image uploads
- Stable performance under normal usage conditions

5.2 Safety Requirements

The system must ensure that data integrity is maintained and that misuse or unauthorized modification of reports is prevented. Submitted evidence should remain unchanged after upload to preserve authenticity and reliability.

The system should prevent accidental deletion or alteration of critical records. Access to sensitive operations must be restricted to authorized users only.

Safety measures include:

- Protection against unauthorized modification of reports
- Preservation of uploaded evidence authenticity
- Prevention of accidental deletion of records
- Controlled access to sensitive system operations

5.3 Security Requirements

Security is essential because the system stores personal data, violation records, and official actions. The system must implement secure authentication and role-based access control to ensure that only authorized users can access specific features.

Sensitive data must be protected during storage and transmission. Secure communication protocols should be used to prevent unauthorized access or data breaches.

Security requirements include:

- Secure user authentication and login system
- Role-based access control for citizens, officers, and administrators
- Protection of personal and violation data
- Secure transmission of data using encrypted communication
- Prevention of unauthorized system access

5.4 Software Quality Attributes

The system should be reliable, user-friendly, and easy to maintain. The interface must be simple enough for users with basic technical knowledge, ensuring accessibility and ease of use.

The design should allow future enhancements without requiring major restructuring. The system should function consistently across supported browsers and devices.

Quality attributes include:

- Reliability and consistent system operation
- Ease of use for non-technical users
- Maintainability for future improvements
- Compatibility across devices and browsers
- Scalability for future feature additions

6. Other Requirements

This section describes additional requirements that support the proper functioning, reliability, and long-term usability of the Traffic Violation Reporting System. These requirements do not fall strictly under functional or nonfunctional categories but are essential for maintaining system integrity, data organization, and operational effectiveness.

The system must maintain structured digital records of users, violation reports, challans, and reward points. Proper data management ensures accountability, traceability, and efficient retrieval of records when required by authorities or administrators. Regular data backup mechanisms should be considered to prevent loss of critical information.

The system should comply with general traffic regulation practices and support digital documentation that aligns with administrative procedures. It must also maintain accurate logs of user activities and system actions to support transparency and monitoring.

To ensure future scalability, the system design should allow additional features such as online challan payment, government database integration, and automated violation detection to be incorporated without major redesign.

Other requirements include:

- Maintenance of structured digital records and logs
- Data backup and recovery considerations
- Support for administrative monitoring and auditing
- Compliance with traffic regulation documentation practices
- Expandability for future system enhancements

Appendix A: Glossary

This section defines important terms and abbreviations used throughout this document to ensure clarity and consistent understanding among readers.

Challan – An official penalty notice issued for violating traffic rules.

Traffic Violation – Any act that breaks traffic laws, such as signal jumping, overspeeding, illegal parking, or riding without safety gear.

GPS (Global Positioning System) – A satellite-based navigation system used to determine location.

RC (Registration Certificate) – Official document that proves vehicle registration with transport authorities.

Pollution Certificate (PUC) – Document certifying that a vehicle meets emission standards.

User Interface (UI) – The visual layout through which users interact with the system.

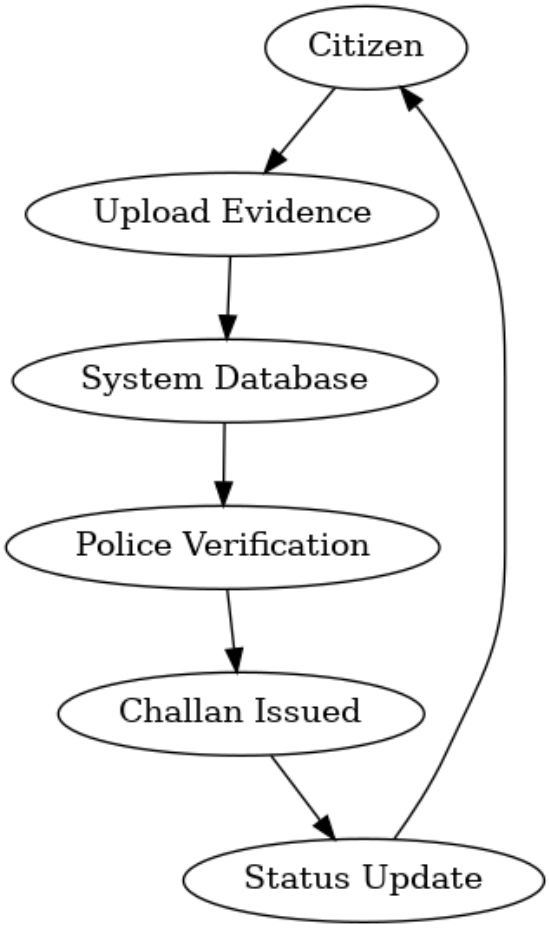
Administrator – Authorized person responsible for managing system settings and user accounts.

Database – Structured storage used to manage and retrieve system data.

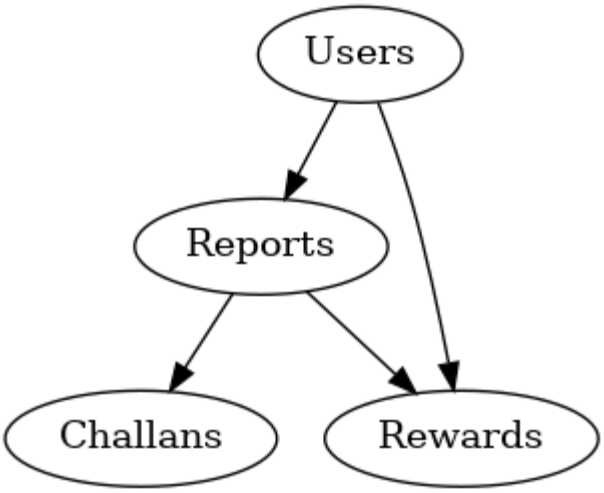
Appendix B: Analysis Models

1. USE CASE DIAGRAM

2.SYSTEM FLOW DIAGRAM



3.ER DIAGRAM



Appendix C: Issues List

This section identifies limitations and considerations related to the current implementation of the system.The academic version of the system does not include live integration with government vehicle databases. Instead, vehicle information and document validity may be stored in a local database for demonstration purposes.Online challan payment functionality is not included in the current version and is considered a future enhancement. Similarly, automated violation detection using artificial intelligence is outside the scope of the initial implementation.The system depends on internet connectivity for real-time reporting and verification. Poor network conditions may affect system responsiveness.

Current limitations include:

- No integration with official government databases
- Online challan payment not implemented
- AI-based violation detection not included
- Dependence on internet connectivity